

FORESIGHT — Demand & Inventory Intelligence

Executive Readout

Estimated Sales at Risk: ■3,452,576

Estimated Capital Locked: ■32,976,851

SKUs Requiring Action: 26

These figures are planning estimates calculated from the current repository's model forecasts, latest inventory snapshot, selling prices and cost prices.

1. Business Problem

FORESIGHT is designed to help an operations team understand expected demand, identify possible stockouts, identify overstock, and decide which SKUs need attention.

The project converts sales, product, calendar and inventory data into forecasts, inventory-risk flags, recommended actions and an interactive dashboard.

2. Data & Workflow

Data sources: daily sales, SKU master, calendar and inventory snapshots.

Workflow: data validation → analysis-ready dataset → EDA → seasonal-naive baseline → Random Forest forecast → risk scoring → decision table → Streamlit dashboard.

3. Forecasting Results

Metric	Current Model Result
MAE	13.48
RMSE	23.17
WAPE	16.85%
Bias	-6.89

The repository also contains SKU-level seasonal-naive WAPE results. No single aggregate baseline WAPE is stored in the model evaluation file, so this readout does not invent a direct aggregate baseline comparison.

4. Inventory Risk

Stockout logic: $\text{lead-time demand} = \text{forecast} \times \text{lead time} \div 7$; $\text{available stock} = \text{current stock} + \text{on-order}$. Reorder is flagged when available stock is below lead-time demand.

Overstock logic: $\text{current stock} > \text{forecast} \times 4$. The $\times 4$ threshold is a project assumption and should be reviewed with business stakeholders.

Current calculation: 5 REORDER, 21 OVERSTOCK, 24 MONITOR.

5. Rupee Impact

Estimated sales at risk: **■3,452,576**

Estimated capital locked in overstock: **■32,976,851**

Sales at risk = forecast demand × selling price for REORDER SKUs. Capital locked = current stock × cost price for OVERSTOCK SKUs.

These are exposure estimates, not realized losses or guaranteed savings. The SKU-level calculations are provided in rupee_impact_analysis.csv.

6. Dashboard

The deployed Streamlit dashboard provides SKU/category/action filters, KPI cards, risk distribution, forecast vs actual, stock vs forecast, reorder priorities, decision table and CSV download.

Live dashboard: foresight-demand-inventory-intelligence-pkcu93wnjwyoxuutcmzueh.streamlit.app

7. Recommendations & Limitations

- Review REORDER SKUs against supplier lead times and purchasing constraints.
- Review OVERSTOCK SKUs for appropriate inventory-clearing actions.
- Refresh forecasts and risk scores when new sales and inventory data become available.
- Forecasts are estimates and can be affected by unusual demand and promotion effects.
- The overstock threshold is rule-based and should be validated with business stakeholders.
- Rupee figures are estimates of exposure, not realized financial outcomes.
- The FastAPI scoring service is not deployed.